# Measure with Cubes

TEACHER KEY

1st Grade Math · Interactive Math · Explore & Reflect

## Get started

Line up cubes end to end to measure a crayon, marker, pencil and ribbon. See why gaps and overlaps give wrong measurements, then compare the lengths. Open the tool and use it to answer each question.

## Explore

1. How many cubes long is the pencil?
   **7 cubes**
2. How many cubes long is the crayon?
   **4 cubes**
3. How many cubes longer is the pencil than the crayon?
   **3 cubes**
4. Put the crayon, marker, pencil and ribbon in order from shortest to longest.
   **crayon (4), marker (6), pencil (7), ribbon (11)**

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## Explore more

**5.** Measure the pencil with gaps. How many cubes reach the end? Is that the real length?

**5 cubes. No — gaps make the count too small; it is really 7.**

**6.** Measure the pencil with overlapping cubes. How many cubes? Is that right?

**11 cubes. No — overlaps make the count too big.**

## Reflect

**7.** Reflect: Why must the cubes touch with no gaps and no overlaps?

**So every part of the length is counted exactly once.**

## Teacher notes

- Model a gap and an overlap on the projector, then ask students to predict whether the count will be too big or too small before pressing +.
- Connect to real cubes: students measure a classroom pencil and compare with the tool.
- Ask comparison questions: How many cubes longer is the ribbon than the marker? (5)

**Standards:** 1.MD.A.1, 1.MD.A.2. Accept any correct answer where several are possible, and ask students to explain with the tool's picture.